



OFFICIAL

Tobacco (Nicotine) – Patient Management Guideline

1. Purpose

The purpose of this document is to provide guidance for the management of patients who may be affected by nicotine dependency in accordance with the [Smoke Free Policy \(health.wa.gov.au\)](https://www.health.wa.gov.au) and [Alcohol and Prohibited Substances Policy](#).

2. Applicability

This guideline applies to all services and staff working across the Armada Kalamunda Group.

3. Key practice points

- Early detection of withdrawal and preventing the risks associated with withdrawal are the key components for effective and safe management – alcohol, tobacco and other drugs screening should occur at the point of presentation or admission to the hospital.
- The smoking and vaping status of all patients must be documented on admission and during their treatment and recorded in the progress notes. Complete the risk assessment, smoking section on the ED triage chart and care plans.
- All patients at risk of experiencing nicotine withdrawal symptoms are to be managed using the ABCD Model of Patient Care.
- Patients, visitors, contractors, and staff must be informed that smoking is prohibited on all hospital grounds and AKG premises. From 28 June 2022, at AKG, adult (18 years of age and over) involuntary mental health inpatients will no longer be exempted from the Smoke Free Policy.

Patients who wish to continue smoking

Key Alert: Staff are not to assist a patient to smoke in courtyards/sites/hospital grounds

Patients suspected or confirmed to have respiratory illnesses (e.g., COVID 19) who smoke must NOT leave their isolation area (beige or red carded) to smoke cigarettes or use e-cigarettes/personal vaporisers.

- Ensure these patients are offered/receive early combination nicotine replacement therapy (NRT), including the faster acting Quickmist® spray (peak time of effect <5 mins) if they are able/have capacity to use this device. Otherwise discuss and provide them with the lozenge or other fast acting options.

Mental Health (MH) Patients

- Vapes, lighters and matches are considered an unsafe item and must be removed from patient belongings (see [AKG Patient Property in Mental Health Wards](#) and EMHS [Searching of Mental Health Patients Belongings and Property](#) for guidance).

Before referencing this guideline document please ensure that you have the latest version from the [Policy Library](#) information hub page.

- It is recommended that voluntary patients voluntarily submit their tobacco supplies, and other tobacco related items are kept in patient storage (staff access only) areas. However, if they choose to keep on their person, they may do so. Involuntary patients will not be permitted to have tobacco products on their person.
- Leave will not be granted for the purpose of smoking however if a patient has day or overnight leave for an approved activity then they are permitted to access and take any smoking products, including lighters with them. Patients will have to return these items to nursing staff if they choose to bring them back. Refer to the [Granting and Management of leave for AKG Mental Health Inpatient Unit Guideline](#).
- In order to prevent environmental tobacco exposure (ETS) for staff, a patient who is identified for leave that is escorted by a clinician will not be permitted to smoke during this activity. Subsequently they will also not be permitted access to any smoking products as outlined above. Escorted Ground Access (EGA)/Unescorted Ground Access (UGA) is a non-smoking activity as the patient remains on hospital grounds.
- Smoking remains prohibited for all patients under the age of 18 years
- **Patients may choose to leave the site to smoke if they are voluntary in status (and over 18 years), have been seen by their treating team, have approved leave, and are considered medically stable. However, they should always have their nicotine dependency assessment completed (i.e. Heaviness Smoking Index (HSI) and recorded and be re-offered NRT prior to leave.**

General health patients

Patients may leave the site if they are medically stable and voluntary in status (and over 18 years), have been seen by their treating team and have no attachments and/or infusions which prevent them from doing so.

- It must be made clear that they are required to smoke off hospital grounds and premises and not in site-based courtyards/gardens/outside areas
- For medically unstable patients, the risks of leaving the site to smoke should be clearly discussed with patient and documented with a management plan in place. Re-offer the options of NRT to manage symptoms of withdrawal during hospital stay. Escalate to Shift Coordinator / MO and escalate Code Black or Blue as relevant
- Where the patient leaves the site for a cigarette/e-cigarette/personal vaporiser there must be documented evidence that:
 - A nicotine dependency assessment has been completed (i.e. HSI)
 - NRT has been discussed and considered in consultation with the patient and/or carer/significant other and documented in the patient integrated notes.
- If patient has decision making capacity but requires assistance to leave the site, offer the options of NRT to manage withdrawal symptoms. **Staff, students and volunteers must not assist the patient off the site to facilitate smoking. Family or friends may assist/ escort the patient off grounds after consultation with staff.**

Cultural considerations:

Aboriginal and culturally and linguistically diverse groups

- Staff should consider using Aboriginal Health Liaison Officers, interpreters and other communication strategies, as necessary – see [Aboriginal Health Liaison Services](#) and the [Interpreter and Translator Guideline](#) for further information.
- Consider other cultural aspects that may affect setting, expectations, family involvement, follow up and other issues.

Smoke Free WA Health System Policy:

- Smoking is not permitted on any Department of Health (DOH) premises and grounds throughout Western Australia: [Smoke Free Policy \(health.wa.gov.au\)](#) MP 0158/21
- Electronic cigarettes (e-cigarettes) and other personal vaporisers for delivery of nicotine or other substances are not permitted to be used in any area where smoking is restricted.

4. Nicotine withdrawals

- Cigarette smoking is a well-established major risk factor for many medical conditions such as diabetes, hypertension, asthma, depression, chronic obstructive pulmonary diseases, emphysema, osteoporosis, and 16 common cancers.¹
- Patients should never be forced to experience nicotine withdrawal symptoms while in hospital. Nicotine withdrawal symptoms are predictable, preventable and treatable.²
- Nicotine withdrawals occur when nicotine levels in the body become low and can start within 30 minutes from when a patient last smoked a cigarette.
- Nicotine has a short half-life of 1-2 hours therefore patients may feel like they need another dose.^{3,4}
- Withdrawal symptoms can cause clinically significant distress, particularly within the first 24 hours when symptoms are most severe.
- The severity and timescale of withdrawal symptoms will vary among patients. However, these withdrawal symptoms fade significantly within two to four weeks within stopping smoking.⁵
- The symptoms of nicotine withdrawal include⁶:
 - Cravings for cigarettes/nicotine
 - Anxiety, irritability, restlessness, agitation, crying, depressed mood
 - Reduced concentration
 - Sleep disturbances / insomnia
 - Headaches
 - Increased appetite
 - Gastrointestinal disturbances
 - Cold symptoms (coughing, sore throat, sneezing, dizziness, mouth ulcers).

Note: Nicotine withdrawal symptoms **are predictable, preventable and treatable**.² Nicotine replacement therapy (NRT) should be repeatedly offered to patients identified as at risk of withdrawal for the duration of their hospital stay. Refer to the ABCD Model of Patient Care section.

5. Nicotine-dependant inpatient management

5.1 Nicotine Replacement Therapy

- NRT has been shown to decrease nicotine withdrawal-induced patient aggression.⁷
- NRT aims to replace the nicotine from smoking cigarettes and e – cigarettes/personal vaporisers with safer alternatives in order to relieve nicotine withdrawal symptoms and reduce cravings for nicotine.⁸
- Insufficient NRT may result in ongoing withdrawal symptoms, which may be mistaken for NRT side effects.
- Transdermal nicotine patches (patch), lozenges, nicotine oral spray (oral spray- Quickmist) are available at AKG. Oral spray works in the buccal mucosa and is not to be inhaled.
- Pharmacological treatment of nicotine addiction withdrawal symptoms in hospitalised smokers potentially improves patient comfort, increases compliance with smoke free policies and promotes smoking cessation after discharge.⁹
- There is abundant evidence that hospital-based programs are highly effective at supporting patients to stop smoking, particularly when pharmacotherapy is provided with counselling, behavioural supports and post-discharge support.¹⁰

5.2 Combination therapy

- Combination therapy is the provision of fast or intermittent acting products (such as oral spray or lozenge) combined with the patch which provides sustained nicotine levels throughout the day.
- Combination therapy is recommended for most people who smoke, especially patients who are highly dependent, unable to stop using one product or experience cravings while using one product alone.¹¹
- Fast acting products should be taken in anticipation of a smoking trigger if possible, and/or when cravings arise.

See [Appendix 1](#) NRT formulations and guidelines for use.

5.3 Nurse/Midwife initiated, and Pharmacist prescribed NRT

- The nurse/midwife may initiate, or pharmacist may prescribe NRT if the Medical Officer (MO) is unavailable. This includes combination therapy for inpatients over 18 years old, following completion of the Heaviness of Smoking Index (HSI) (see Table 1 and refer to the ABCD Model of Patient care section).
 - Nurse/midwife: Complete prescription of NRT on “Once only, pre-medication and nurse/midwife-initiated medicines” section of the Hospital Medication Chart (HMC) and sign the entry.
 - Pharmacist: Complete prescription of NRT on “Regular and prn” section of the Hospital Medication Chart (HMC) and sign the entry. Pharmacist to annotate on chart for review of the second dose.

- Ensure the MO reviews the patient and prescribes ongoing NRT (within 24 hours of nurse/midwife initiated and pharmacist prescribed dose).
- If a MO is not available to review the patient a phone order must be received before further NRT can be administered to the patient.
 - Assessment of nicotine dependence using HSI must be documented routinely for all patients (within 30 minutes of presentation, where feasible/practicable).
- NRT form to remain in patient's bedside/integrated notes.
- Nurse/midwife can now initiate multiple doses of combination NRT within the first ≤ 24 hours, following this time period NRT is to be prescribed by MO and reviewed by pharmacist.

5.4 Nurse/midwife/pharmacist excluded patients

- NRT must **not** be initiated/prescribed by nursing/midwifery/pharmacist staff for patients under the age of 18. A MO will prescribe NRT in these circumstances.
- For patients with either suspected or confirmed respiratory illness (e.g. COVID 19), application of the type of NRT must be made in consultation with a MO.

Table 1 Exclusions from nurse/midwife initiated and pharmacist prescribed NRT

Contraindicated	Precaution (MO consultation required)
● Non-smokers ● Children aged < 12 years ● Nicotine hypersensitivity ● Phenylketonurics (should not use lozenge) ● Menthol hypersensitivity (should not use oral spray) ● Buerger's Disease (Thromboangiitis obliterans)	● Persons aged < 18 years ● Acute myocardial infarction, unstable or worsening angina, severe cardiac arrhythmias ● Peripheral vascular disease with critical limb ischaemia ● Recent* cerebrovascular accident

* Recent is defined as within the last four weeks

5.5 NRT in Pregnancy

- If the patient is already using NRT on admission, nursing/midwifery staff may continue to provide it. A MO is to review and prescribe within 24 hours.
- NRT can be offered at any stage of pregnancy or in the postnatal period if the mother has tried to stop without success or wants to stop and the benefits of stopping outweigh the risk of NRT and continued smoking.¹²
- Intermittent use NRT products are preferred for pregnant and breastfeeding women because they provide smaller daily doses of nicotine than patches.¹²
- A patch can be used if oral NRT is not tolerated or if more NRT is required to manage withdrawals (i.e. combination therapy).
- Pregnant women using patches should remove patches at night to allow an eight-hour break and apply a new patch in the morning.¹²

5.6 NRT Administration Precautions

- The oral spray is not specifically required to be sprayed sublingually or buccally but rather into the mouth. Patients should be advised to avoid directly spraying onto the tongue or into

the throat. All patients should be advised to avoid inhaling the spray as that is not the intended site of administration.

- Following the HSI nicotine dependence assessment, if the patient's preference of fast acting NRT options has been identified as the oral (Quickmist®) spray, this is to be documented on the medication chart.
- Once charted and the patient educated on NRT use/dosage, patients are to be assessed for risk to self and others with regards to keeping the oral (Quickmist®) spray on their person/by the bedside. If this is deemed unsafe, the spray will be labelled and stored in the patient's locked medication drawer/trolley, or the patient will be offered another form of fast-acting NRT.
- Encourage the patient to inform the nurse when the oral (Quickmist®) spray is self-administered. This is to ensure efficacy of the product and documentation of the dose/use and time on the medication chart.
- Patients with chronic throat conditions or unstable asthma to use the oral spray with caution.
- Nicotine patches should not be used on patients with adverse reactions to adhesive dressings or on areas of inflamed or broken skin (extreme caution in patients with generalised skin conditions such as psoriasis, eczema, dermatitis etc.).
- NRT patches should be applied to clean, dry and non-hairy skin (with application site rotated each time the patch is changed). Avoid using oral sprays in patients who are sensitive to menthol.
- Patches contain aluminium and must be removed prior to any magnetic resonance imaging (MRI).
- Patches should remain in situ if the patient is going for a surgical procedure. Ensure the patch is not positioned near site of surgery or application of equipment (e.g. ECG electrodes). If removed, patients may wake in recovery in withdrawal due to low serum nicotine concentrations.
- Patients aged over 64 years and/or patients with dementia and other cognitive impairments and those on soft diet are to be assessed as to their ability to use lozenges appropriately.
- Nursing/midwifery/pharmacy staff will consult a MO before initiating NRT for patients who are pregnant or breast feeding; or who have had recent cardiovascular events (e.g. recent myocardial infarction, recent cerebrovascular accident See [Appendix 2](#) Common drug interactions with smoking cessation.

5.7 Patients of Concern

- Patients with suspected or confirmed respiratory illnesses (e.g., COVID 19) or other highly infectious conditions must remain in their isolation environment and are not permitted to leave for smoking, e-cigarette use, or personal vaporiser use.
- If patient is a significant risk to themselves or to others, offer the option of NRT to manage symptoms of withdrawal (offer a different type of NRT if current NRT is not effective). These patients may not leave the ward.
- The risks of leaving the ward to smoke should be discussed with medically unstable patients. In some situations, patients will have an immediate need for nicotine and may experience high anxiety and problematic behaviour when denied a cigarette:
 - Offer fast acting NRT (e.g. oral spray or lozenge).
 - Escalate to MO/Drug and Alcohol Clinical Nurse Consultant to review patient if anxiety, irritability, restlessness/agitation increase.

- Commence Constant Observation Chart as applicable.
- Notify Security as applicable.
- For patients with diminished capacity (e.g. sedated) NRT may be applied to help reduce symptoms of withdrawal prior to de-sedation.

5.8 Other pharmacotherapies

Other pharmacotherapies (i.e. Champix and Zyban) for smoking cessation that are not NRT may only be prescribed to patients by the treating doctor.

6. ABCD Model of Patient Care

The ABCD Model should be delivered to all patients who smoke, regardless of their intention to stop smoking.

Table 2 ABCD Model of Practice

A	Ask	● Ask every patient (where clinically possible) about their smoking status (current smoker, ex-smoker, never smoked) and/or electronic cigarette (e-cigarette) use. ● Record smoking status on the Medical Assessment Form or equivalent. For patients identified as smokers, complete the Heaviness of Smoking Index (HSI) assessment on the NRT Management & Withdrawal Chart AKMR27.1. Keep the NRT chart with the medication chart. ● Patients should be assessed for nicotine dependence and withdrawal within 30 minutes of admission. Document withdrawal symptoms as part of their treatment plan (nicotine has a short half-life of 1 - 2 hours: therefore, patients who smoke may soon feel like they need another dose)
B	Brief advice	● Use positive messaging to patients who smoke ● Reassure patients that we will work with them to regularly assess and manage their nicotine withdrawal symptoms while in hospital through NRT and behavioural interventions. ● Ask the patient if they have previously used NRT and if so, what NRT has worked for them. If they say NRT did not work, explain this may have been due to under-dosing. Discuss the range of options and effectiveness of combination therapy. ● Advise patient how to use intermittent (fast acting) formulations to manage/reduce cravings. Refer to Appendix 1 - NRT formulations and directions for use. ● Encourage all eligible patients who smoke (includes e- cigarette/personal vapouriser users) to use NRT to abstain from smoking whilst in a smoke-free hospital setting. ● Offer or re-offer NRT throughout the patient's stay. ● NRT can be Nurse or Midwife initiated, or Medical Officer or Pharmacist prescribed. ● Chart NRT dosage as per the patient's HSI score.

		- Review and document withdrawal symptoms in NRT Management & Withdrawal Chart AKMR27.1. - Advise all smokers, in a clear, non-confrontational and non-judgmental way, to stop smoking or cut down using NRT. - Offer Quit pack, ward based behavioural resources 'e.g. are you craving a cigarette booklet' and referrals to relevant supports e.g. Quitline Mooditj Walyun (meaning strong lungs), Resource Packs are also available for Aboriginal patients.
C D	Communication at discharge	- Assess the patient's intention to reduce or stop smoking after discharge. Use of pharmacotherapy alone may not be sufficient, therefore provide behavioural intervention and supports upon discharge to the home and community i.e. behavioural resources, Quitpack and referral to Quitline. Patients who smoke need support to manage cues, habits and behaviours related to smoking if long term success is to be achieved. - Re-offer Quit pack and referrals to relevant supports if applicable e.g. Quitline and/or YIU Resource Packs. - Include smoking status, support received and plans for follow-up in the Discharge Summary (NaCS) and/or discharge envelope.

6.1 Ask – Assessment documentation

- Complete Medical Assessment Form
- Complete the HSI on the NRT Management & Withdrawal Chart AKMR27.1
- Smoking status and NRT must be raised with patients on attendance to pre-admission clinics.
- Ideally, patients should be assessed for nicotine dependence and withdrawal within 30 minutes of admission.

Table 3 Heaviness of Smoking Index

How soon after waking do you have your first cigarette? (Circle one response)		Within 5 min	3
		6-30 min	2
		31-60 min	1
		After 60 min	0
How many cigarettes do you smoke each day? (Circle one response)		10 or less	0
		11 – 20	1
		21 – 30	2
		31 or more	3
(Responses from these two questions are summed and a score from 0 to 6 results. A higher score represents higher dependence).			
Total Score			
Dependence Score	0-1 = low 2-4 = moderate 5-6 = high		

6.2 Brief advice – practice guidance

Offer appropriate level of NRT according to their level of dependence as per their HSI score:

Table 4 Combination Therapy NRT Dosage

Dependence Level	Combination Therapy NRT Dosage
High or Moderate	Patches: 21mg/24hr AND Oral spray: 1mg/spray – 1-2 sprays into the mouth, up to a Maximum 4 sprays/hr. and 32 sprays/24hrs when used with patch (max. 64 sprays/24hrs without patch) OR Lozenge: 2mg – Maximum 12 lozenges/24hrs when used with patch (max. 15 lozenges/24hrs without patch)
Low	Patches: 14mg/24hr AND Oral spray: 1mg/spray – 1-2 sprays into the mouth, up to a Maximum 4 sprays/hr. and 32 sprays/24hrs when used with patch (max. 64 sprays/24hrs without patch) OR Lozenge: 2mg – Maximum 12 lozenges/24hrs when used with patch (max. 15 lozenges/24hrs without patch).

Nicotine Management and Withdrawal Plan:

- Document withdrawals on the NRT Management & Withdrawal Chart AKMR27.1
- A review is necessary of all patients who are using NRT and still experiencing cravings.
- Patients who refuse NRT should be closely monitored for nicotine withdrawal symptoms and reoffered NRT and reminded that NRT is available during their hospital stay.
- Discuss patient's previous experience of using NRT, if applicable. If patients say "NRT did not work" explain this may have been due to under-dosing and combination therapy is effective.

6.3 Communication at discharge – practice guidance

- Provide a copy of the NRT Management & Withdrawal Chart AKMR27.1 with the discharge summary.
- Assess the patient's intention to remain abstinent after discharge. If the patient is ready to cease smoking:
 - The pharmacotherapy treatment must be the sole PBS-subsidised therapy for this condition.
 - Advise the patient that they only have access to PBS subsidised funding of one form of NRT at a time.
 - Patient must be undergoing concurrent counselling for smoking cessation through a comprehensive support and counselling program e.g. Quitline or be about to enter such a program at the time PBS-subsidised treatment is initiated.
- AKG patients are charged for all discharge medications that are dispensed at the hospital.
- If patients take the prescription(s) to an external/community pharmacy, they will need to pay for their own medications:

- PBS entitlements change depending on whether the patient is a repatriate, Aboriginal Torres Strait Islander, non-Aboriginal Torres Strait Islander, or from Reciprocal Health Care.
- Counsel patient that certain forms of NRT are subsidised by the PBS and that they should consult their GP or pharmacist for further information.
- Quit and Mooditj Walyun packs can be obtained through the Ward CNS and supplied on each ward as requested.

7. Staff education and resources

All clinicians should have the capacity to provide brief intervention on nicotine dependence and cessation support to patients. Refer to the following resources:

- Take 5 PowerPoint (currently under review, will be available on the EMHS Smoke Free intranet hub page)
- [Smoke Free EMHS](#) (Intranet hub page)
- [Brief advice: Ask, Advise, Help](#) (external learning package videos)
- [Quit Learning Hub](#) (external self-directed learning package)
- [Registration Troubleshooting Cheat Sheet](#) (Chrome is recommended)
- [EMHS Smoking Care for Inpatients – Dr Kelly Ridley](#)
- [Tobacco in Australia \(Facts and Issues\)](#)
- [Management of Nicotine Dependency across EMHS](#) (internal learning package)

8. Support for staff who smoke

East Metropolitan Health Service (EMHS) staff can access a twelve-week course, per year of Nicotine Replacement Therapy (NRT), free of charge, to help stop smoking or manage withdrawals during work hours.

To access, staff need to:

- Complete the Employee Application for NRT housed in the [Support for Staff who Smoke](#) section of the EMHS Smoke Free hub page
- Hand in the completed form to the Pharmacy Department.

9. Legislative context

The following documents are required to give effect to this policy:

- HDWA
 - [Smoke Free Policy](#)
- EMHS
 - [Smoke-free Environment Policy EMHS: 124](#)
 - [Alcohol and Prohibited Substances Policy EMHS: 112](#)

10. Supporting information

The following documents support the implementation of this policy:

- AKMR27.1 Nicotine Replacement Therapy (NRT) Management and Withdrawal Chart
- [Guidelines for the implementation of the smoke free policy](#)

- [Patient Property in Mental Health Wards Procedure](#)
- [Searching of Patients Belongings and Property](#)
- [Granting and Management of Leave for Mental Health Inpatients Policy](#)
- [Interpreter and Translator Guideline](#)
- [Aboriginal Health Liaison Service](#)

11. Compliance, monitoring and evaluation

Compliance with this document will be evaluated by the relevant departmental staff via routine clinical incident review processes.

12. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 4 – Action 4.3 Partnering with consumers
- Standard 4 – Action 4.10 Medication review
- Standard 5 – Action 5.10: Clinicians use relevant screening processes:
- Standard 5 – Action 5.13: Clinicians use processes for shared decision making to develop and document a comprehensive and individualised plan that:
- Standard 5 – Action 5.34: The health service organisation has processes to support collaboration with patients, carers and families

13. References

1. The Royal Australian College of General Practitioners. (2019 December) Supporting smoking cessation: A guide for health professionals. Second edition [Internet]. <https://www.racgp.org.au/clinical-resources/clinical-guidelines/key-racgp-guidelines/view-all-racgp-guidelines/supporting-smoking-cessation/introduction-to-smoking-cessation#ref-num-19>
2. Dean, E. (2018, May). Tackling tobacco in mental health settings [presentation]. Alfred Health. <https://cancerwa.asn.au/wp-content/uploads/2022/07/2018-06-21-Tackling-Tobacco-in-Mental-Health-Settings-Emma-Dean.pdf>
3. National Center for Biotechnology Information (2022). PubChem Compound Summary for CID 89594, Nicotine. <https://pubchem.ncbi.nlm.nih.gov/compound/Nicotine>.
4. Dresden D. (2020, January). How long does nicotine stay in your system? [Internet]. Medicalnewstoday.com. <https://www.medicalnewstoday.com/articles/322526>
5. Benowitz, N. L. (2009). Pharmacology of Nicotine: Addiction, Smoking-Induced Disease, and Therapeutics. Annual Review of Pharmacology and Toxicology, 49, 57-71. <https://doi.org/10.1146/annurev.pharmtox.48.113006.094742>
6. Alfred Health. (2017). Pharmacotherapy for smoking cessation. <https://www.alfredhealth.org.au/contents/resources/community-and-health-promo/Pharmacotherapy-for-Smoking-Cessation.pdf>.
7. Make Smoking History. (n.d.). Managing Withdrawal Symptoms. <https://www.makesmokinghistory.org.au/quit-support/support-after-quitting/withdrawal-symptoms>

8. Lawn, S. & R. Pols. (2005). Smoking bans in psychiatric inpatient settings? A review of the research. Australian and New Zealand Journal of Psychiatry, 39(10), 866-885.
<https://doi.org/10.1080/j.1440-1614.2005.01697.x>
9. Greenhalgh, EM., Dean, E., Stillman, S., & Ford, C. 7.16 Pharmacotherapies for smoking cessation. In Greenhalgh, EM, Scollo, MM and Winstanley, MH [editors]. Tobacco in Australia; Facts and issues. Melbourne; Cancer Council Victoria; 2021. Available from: <http://www.tobaccoinaustralia.org.au/chapter-7-cessation/7-16-pharmacotherapy>
10. Mullen, K., Manuet, DG., Hawkin, SJ., Pipe., Coyle, D., Hobler, LA., Younger, J., Wells, GA., & Reid, RD. 2017. Effectiveness of a hospital-initiated smoking cessation programme: two-year and healthcare outcomes.
<https://tobaccocontrol.bmj.com/content/tobaccocontrol/26/3/293.full.pdf>
11. Nicotine Dependence. Australian Medicines Handbook. (2022). Adelaide: Australian Medicines Handbook Pty Ltd. Retrieved from <https://amhonline.amh.net.au/>
12. Quit for new life NRT Protocol. (2017 February).
https://www.guild.org.au/__data/assets/pdf_file/0018/14517/protocol-for-the-provision-of-nicotine-replacement-therapy-in-qfnl-feb-2017.pdf.
13. Women and Newborn Health Service. (February 2025). Guideline: Management of Nicotine Dependence. Perth, WA.

14. Document owner

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15. Review

This guideline will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
V2.2	22 01 2026	06 11 2026	Inclusion of vaping and changes to Appendix 3 flow chart

16. Authorisation

This guideline has been authorised and issued by the:

Authorised by	Alisha Thompson - Executive Director AKG
Authorisation date	22 01 2026
Published date	22 01 2026

Appendix 1 – NRT formulations and directions for use

	Nicotine dependency level (HSI score)	NRT formulation	Dosage	Directions	Precautions
SLOW ACTING	Moderate-High (>10 cigarettes daily) and patient weighs >45kg	Patches (21mg)	21mg/24hours	Apply patch to a clean, dry, hairless area of skin. Rotate application site each day, avoiding the same site for 3 days to avoid skin irritation. Time to peak: 6-10 hours.	Avoid applying patch to areas of sensitive skin. Exercise caution using patches in patients with psoriasis, dermatitis, eczema and other inflammatory skin conditions. Contains aluminium - must be removed prior to MRI.
	Low (≤10 cigarettes daily), and/or patient weighs <45kg	Patches (14mg)	14mg/24hours OR use immediate release NRT formulation alone.		
FAST ACTING	All HSI scores alone or in combination with patch	Oral Spray (1mg/spray)	1-2 sprays when cigarette would normally be smoked or cravings emerge. Maximum 4 sprays/hour (64 sprays/day) if alone; Maximum 32 sprays/day if used in combination with patch	Spray 1 spray into the open mouth, avoiding the lips. Do not inhale the spray, and avoid swallowing for a few seconds after dosing. Avoid food/drinks while dosing the spray. Time to peak: <5 minutes.	Avoid in patients with inflammatory mouth/throat conditions. Take care not to spray near the eyes.
	All HSI scores (first cigarette ≤30 minutes after waking)	Lozenges (2mg)	4mg every 1-2 hours. Manufacturer does not recommend use with patch	Place lozenge in the mouth and move from one side to the other periodically until completely dissolved (20-30 minutes). Do not chew or swallow the lozenge. Avoid food/drink while the lozenge is in the mouth.	Avoid in patients with inflammatory mouth/throat conditions. Swallowed nicotine may exacerbate oesophagitis and gastric/peptic ulcers. Contraindicated in patients with phenylketonuria.
	All HSI scores (first cigarette > 30 minutes after waking) OR when used in combination with patch		2mg every 1-2 hours. Only use monotherapy if low dependence, otherwise use in combination with patch	Max dosage of any strength is 15 lozenges/24 hours (or 12 if using in combination with patches). Time to peak: 20-30 minutes.	
	All HSI scores OR used in combination with patch	Inhalator (15mg/cartridge)	Patient to self-titrate dosage according to cravings/withdrawal symptoms; 1 cartridge = 7 cigarettes. Maximum 6 cartridges per 24 hours	Open inhalator device, insert cartridge and reseat inhalator device. Inhale in the same way as a cigarette. One cartridge generally lasts 40 minutes of intensive use. Time to peak: 30 minutes.	Use with caution in patients with chronic throat conditions or respiratory disease. Avoid in patients with menthol hypersensitivity.
GENERAL NRT PRECAUTIONS: NRT should not be prescribed for patients <12 years or those with Buerger's disease (Thromboangiitis obliterans). Patients with unstable cardiovascular disease (MI, angina, arrhythmia, CVA) and pregnant patients should attempt smoking cessation without NRT if possible – consult medical staff if NRT is required. Patients with diabetes should monitor BGLs more frequently. Clearance of nicotine may be reduced in those with hepatic/renal impairment and in the elderly; monitor for nicotine adverse effects.					

Appendix 2 – Common drug interactions with smoking cessation

Drug	Mechanism
Cardiovascular drugs:	
Propranolol	Smoking increases clearance. Closely monitor for adverse effects.
Verapamil	Smoking increases clearance. Closely monitor dose
Warfarin	Dose reduction of 14-23% needed. Closely monitor INR
Mexiletine, flecainide, lignocaine	Dosage may need to be decreased
Benzodiazepines	
Diazepam	Smoking increases clearance. Monitor for increased sedation post cessation of smoking.
Other Benzodiazepines	Smoking may increase clearance. Monitor for increased sedation.
Antipsychotics	
Clozapine	Smoking increases clearance. Dose reduction may be needed to avoid toxicity.
Olanzapine	Smoking increases clearance. Monitor.
Haloperidol	Smoking increases clearance. Monitor.
Antidepressants	
Fluvoxamine	Smoking increases clearance. Monitor for adverse events post smoking cessation.
Tricyclic antidepressants	Smoking may increase clearance. Monitor.
Alzheimer's	
Rivastigmine	Smoking increases clearance. Decreased dose may be needed.
Tacrine	Smoking increases clearance. Decreased dose may be needed.
Antidiabetic	
Insulin	Smoking may reduce subcutaneous insulin absorption. Post smoking Cessation monitor BSLs. May need dose reduction.
Oral Hypoglycaemics	Nicotine may increase plasma glucose. Monitor BSLs. May need dose reduction.
Respiratory	
Theophylline	Decrease in clearance after smoking cessation. Closely monitor levels and adjust dose accordingly.
Other:	
Caffeine	Increased caffeine levels post smoking cessation. Recommend reduced caffeine intake post smoking cessation.

Appendix 3: Offering Nicotine Replacement Therapy (NRT) for Patients Using Tobacco Products or Dual Users (Patients Who Concurrently Smoke and Vape)

ASK: Assess smoking status by asking if the patient smokes or vapes, and document.

If YES, complete the HSI.



Heaviness of Smoking Index (HSI)		
Q1. How soon after waking do you have your first cigarette?	Within 5 min	☐ 3
	5-30 min	☐ 2
	31-60 min	☐ 1
	60+ min	☐ 0
Q2. How many cigarettes do you smoke each day?"	10 or less	☐ 0
	11 – 20	☐ 1
	21 – 30	☐ 2
	31 or more	☐ 3
TOTAL SCORE		=
DEPENDENCE SCORE LOW = 0–2 MODERATE = 3-4 HIGH = 5-6		

ADVISE: all patients who smoke (and vape) that the service is completely smoke and vape-free, they will most likely go into withdrawal and that NRT is available. Advise them to quit in a clear, non-confrontational and personalised way.

HELP:

- By prescribing (or helping patients to access) combination NRT (**nicotine patch and faster-acting NRT e.g. gum, lozenge, oral spray**) based on HSI Score.
- By offering referral to behavioural intervention through Quitline (13 7848).

LOW: HSI score 0-2 or mild cravings with previous quit attempts

NRT may not be required.

Offer if needed:

Nicotine 2mg chewing gum

1 piece of gum to be chewed as directed PRN up to every 1-2 hours (Max. 12 pieces in 24 hours) (Avoid using >1 piece/hour)

OR

Nicotine 2mg lozenges

1 lozenge to be used as directed PRN up to every 1-2 hours (Max. 15 lozenges in 24 hours)

OR

Nicotine QuickMist Oral Spray 1mg

1-2 sprays as directed PRN

Do not use more than 2 sprays at a time, 4 sprays per hour for 16 hours or 64 sprays per day

MODERATE: HSI score 3-4 or significant cravings with previous quit attempts

Combination NRT is needed.

Nicotine patch 21mg/24 hour

PLUS

Nicotine 2mg chewing gum

1 piece of gum to be chewed as directed PRN up to every 1-2 hours (Max. 12 pieces in 24 hours) (Avoid using >1 piece/hour)

OR

Nicotine 2mg lozenges

1 lozenge to be used as directed PRN up to every 1-2 hours (Max. 15 lozenges in 24 hours)

OR

Nicotine QuickMist Oral Spray 1mg

1-2 sprays as directed PRN

Do not use more than 2 sprays at a time, 4 sprays per hour for 16 hours or 64 sprays per day

HIGH: HSI score 5-6 or significant cravings with previous quit attempts

Combination NRT is needed.

Nicotine patch 21mg/24 hour

PLUS

Nicotine 4mg chewing gum

1 piece of gum to be chewed as directed PRN up to every 1-2 hours (Max. 10 pieces in 24 hours) (Avoid using >1 piece/hour)

OR

Nicotine 4mg lozenges

1 lozenge to be used as directed PRN up to every 1-2 hours (Max. 15 lozenges in 24 hours)

OR

Nicotine QuickMist Oral Spray 1mg

1-2 sprays as directed PRN

Do not use more than 2 sprays at a time, 4 sprays per hour for 16 hours or 64 sprays per day

Ref: QUIT Health Services Clinical Guideline and Pathway Template May 2021

The Heaviness of smoking index (HSI) is a validated tool to assess level of nicotine dependency from smoking only (combustible tobacco). There is currently no validated tool to assess nicotine dependency from nicotine vaping products.

Offering Nicotine Replacement Therapy (NRT) for Those Who Use Nicotine Vaping Products (e-cigarettes/vapes)

Precaution: this prescribing guidance is for current use of Nicotine Vaping Products only

ASK: Time to first vape

ADVISE: all patients who use e-cigarettes that the service is completely Smoke-Free (including e-cigarettes), they will most likely go into withdrawal and that NRT is available. Advise them to quit in a clear, non-confrontational and personalised way.

HELP:

- By prescribing (or helping patients to access) NRT based on ‘time to first vape’.
- By offering referral to behavioural intervention through Quitline (13 7848).

Time to first vape > 30 minutes

Nicotine 2mg chewing gum:
1 piece of gum to be chewed as directed
PRN up to every 1-2 hours
(Maximum 12 pieces in 24 hours)
(Avoid using >1 piece/ hr)
OR
Nicotine 2mg lozenges
1 lozenge to be used as directed PRN up to every 1-2 hrs
(Maximum 15 lozenges in 24 hours)
OR
Nicotine QuickMist Oral Spray 1mg
1-2 sprays as directed PRN
Do not use more than 2 sprays at a time, 4 sprays per hour for 16 hours or 64 sprays per day

Time to first vape < 30 minutes

Combination NRT is recommended
Nicotine patch 21mg/24 hour
PLUS
Nicotine 4mg chewing gum:
1 piece of gum to be chewed as directed
PRN up to every 1-2 hours
(Maximum 10 pieces in 24 hours)
(Avoid using >1 piece/ hr)
OR
Nicotine 4mg lozenges
1 lozenge to be used as directed PRN up to every 1- 2hrs
(Maximum 15 lozenges in 24 hours)
OR
Nicotine QuickMist Oral Spray 1mg
1-2 sprays as directed PRN
Do not use more than 2 sprays at a time, 4 sprays per hour for 16 hours or 64 sprays per day

Ref: Alfred Health Clinical Management of Nicotine Dependency Among Patients 2020, Bittoun, R. (2021). Managing vaping cessation: A monograph for counselling adult and adolescent vapers.

Currently, there is no validated assessment tool to assess nicotine dependency from the use of nicotine vaping products. Regular review is recommended to adjust NRT as needed.

- Notes:**
- 1) Consider commencing at higher dosages (including combination NRT) if the patient has experienced severe cravings when previously abstaining.
 - 2) Nicotine dependence from the use of nicotine vaping products is influenced by nicotine concentration, inhalation technique and vaping device characteristics. The dose of NRT can be increased if the patient has inadequate relief of nicotine withdrawal symptoms (e.g. urges to smoke/vape, irritability and restlessness).
 - 3) ‘Dual users’ (patients who concurrently smoke and vape) can be initiated on NRT as per the patient’s HSI score. For prescribing guidance.
 - 4) Check the patient’s current medication regimen for any potential drug interactions.